Hints: Inequalities – 1. The Cauchy–Bunyakovsky inequality as a technique

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. The functions 1 and x are orthogonal in $L^2[-1,1]$, and $\int_{-1}^{1} 1 = 2$, $\int_{-1}^{1} x^2 = \frac{2}{3}$. So $e_0 = \frac{1}{\sqrt{2}}$, $e_1 = \sqrt{\frac{3}{2}}\,x$ are orthonormal and the right-hand side is $\langle f, e_0\rangle^2 + \langle f, e_1\rangle^2$: this is Bessel's inequality.

Hint 2. Let $g = \langle f, e_0\rangle e_0 + \langle f, e_1\rangle e_1 = \frac{1}{2}\int_{-1}^{1} f + \frac{3}{2}\big(\int_{-1}^{1} t f(t)\,dt\big)x$. Then $f - g$ is orthogonal to 1 and x, hence to g, so $\int f^2 = \int g^2 + \int (f-g)^2 \geqslant \int g^2 = \frac{1}{2}\big(\int f\big)^2 + \frac{3}{2}\big(\int xf\big)^2$. Equality iff $f = g$, i.e. exactly for the linear functions $f(x) = a + bx$.

Problem 2

Hint 1. Put $g(x) = \int_0^1 f(x,y)\,dy$, $h(y) = \int_0^1 f(x,y)\,dx$ and $I = \int_0^1\int_0^1 f$. The inequality reads $\int h^2 + \int g^2 \leqslant I^2 + \iint f^2$. Look for a square whose integral is the difference.

Hint 2. Expand $\iint \big(f(x,y) - g(x) - h(y) + I\big)^2\,dx\,dy \geqslant 0$. Every cross term reduces to $\int g^2$, $\int h^2$ or I^2 (e.g. $\iint fg = \int g^2$, $\iint gh = I^2$), and the total is exactly $\iint f^2 - \int g^2 - \int h^2 + I^2$.

Problem 3

Hint 1. First show that $\mathrm{Var}(f) = \min_c \int_0^1 (f-c)^2$: expand $\int (f - \mu(f) + \mu(f) - c)^2$; the cross term vanishes.

Hint 2. Hence $\mathrm{Var}(fg) \leqslant \int_0^1 (fg - \mu(f)\mu(g))^2$. Split $fg - \mu(f)\mu(g) = (f-\mu(f))g + (g-\mu(g))\mu(f)$.

Hint 3. Use $(u+v)^2 \leqslant 2u^2 + 2v^2$ pointwise, then $|g| \leqslant M(g)$ and $|\mu(f)| \leqslant M(f)$.

Problem 4

Hint 1. Split $\int_0^1 f = \int_0^{1/2} f + \int_{1/2}^1 f$ and integrate each part by parts using $f(\frac{1}{2}) = 0$, with x on the left half and $x - 1$ on the right half.

Hint 2. You get $\int_0^{1/2} f = -\int_0^{1/2} xf'$ and $\int_{1/2}^1 f = -\int_{1/2}^1 (x-1)f'$, i.e. $\int_0^1 f = -\int_0^1 k(x)f'(x)\,dx$ with $k(x) = x$ on $[0, \frac{1}{2}]$ and $x - 1$ on $(\frac{1}{2}, 1]$.

Hint 3. Apply Cauchy–Bunyakovsky: $\big(\int_0^1 f\big)^2 \leqslant \int_0^1 k^2 \cdot \int_0^1 (f')^2$ and $\int_0^1 k^2 = 2\int_0^{1/2} x^2\,dx = \frac{1}{12}$.

Problem 5

Hint 1. Integrate $\int_0^1 xf$ by parts with $f(1) = 0$ so that xf' appears.

Hint 2. $\int_0^1 xf\,dx = \big[\frac{x^2}{2}f\big]_0^1 - \frac{1}{2}\int_0^1 x^2 f'\,dx$, so $2\int_0^1 xf = -\int_0^1 x \cdot xf'(x)\,dx$.

Hint 3. Cauchy–Bunyakovsky with $\int_0^1 x^2 = \frac{1}{3}$: $4\big(\int xf\big)^2 \leqslant \frac{1}{3}\int (xf')^2$. Compare with Problem 4: there $\int f = -\int kf'$ with the kernel k = signed distance to the zero $\frac{1}{2}$ and $\int k^2 = \frac{1}{12}$; here the zero is at 1, the derivative carries the weight x, and the constant $12 = 4 \cdot 3$ comes from the factor 2 of the integration by parts and $\int x^2 = \frac{1}{3}$.

Problem 6

(a)

Hint 1. Put $S = \sum_{j=1}^{n} \frac{x_j}{\sqrt{j}}$ and write $x_i^2 = \frac{x_i}{\sqrt{i}} \cdot x_i\sqrt{i}$. Since the sequence decreases, show that $x_i\sqrt{i} \leqslant S$ for every $i \leqslant n$.

Hint 2. For $j \leqslant i$ we have $x_j \geqslant x_i$ and $\frac{1}{\sqrt{j}} \geqslant \frac{1}{\sqrt{i}}$, so $x_i\sqrt{i} = \sum_{j=1}^{i} \frac{x_i}{\sqrt{i}} \leqslant \sum_{j=1}^{i} \frac{x_j}{\sqrt{j}} \leqslant S$. Hence $\sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} \frac{x_i}{\sqrt{i}} \cdot x_i\sqrt{i} \leqslant S \cdot S$.

(b)

Hint 1. Apply (a) (with $n \to \infty$) to the tail sequence $x_m, x_{m+1}, \ldots$: $\bigl(\sum_{i \geqslant m} x_i^2\bigr)^{1/2} \leqslant \sum_{i \geqslant m} \frac{x_i}{\sqrt{i-m+1}}$.

Hint 2. Swap the sums: the left side is at most $\sum_i x_i \sum_{m=1}^{i} \frac{1}{\sqrt{m(i+1-m)}}$. The inner sum is bounded: by symmetry it is at most $2\sqrt{\frac{2}{i+1}} \sum_{m \leqslant (i+1)/2} \frac{1}{\sqrt{m}} \leqslant 4$ (it is a Riemann sum for $\int_0^1 \frac{dt}{\sqrt{t(1-t)}} = \pi$). So $C = 4$ works.

Problem 7

Hint 1. Write the product as a double integral: $\int e^f \int e^{-f} = \int_0^1 \int_0^1 e^{f(x)-f(y)}\,dx\,dy$, and symmetrise in $x \leftrightarrow y$.

Hint 2. This equals $\iint \cosh\bigl(f(x) - f(y)\bigr)$, and $\cosh t \geqslant 1 + \frac{t^2}{2}$. Finally $\frac{1}{2}\iint (f(x) - f(y))^2 = \int f^2 - \bigl(\int f\bigr)^2$.

Problem 8

Hint 1. The point is that $\int_0^T \ln f(x)\,dx = \int_0^T \ln f(x+\alpha)\,dx$ by periodicity.

Hint 2. Use $t \geqslant 1 + \ln t$ with $t = \frac{f(x)}{f(x+\alpha)}$ and integrate over a period: $\int_0^T \frac{f(x)}{f(x+\alpha)} \geqslant T + \int_0^T \bigl(\ln f(x) - \ln f(x+\alpha)\bigr) = T$.

Problem 9

Hint 1. For $v_1, \ldots, v_m$ and real c_i, $\sum_{i,j} c_i c_j \langle v_i, v_j\rangle = \bigl\|\sum c_i v_i\bigr\|^2 \geqslant 0$: the Gram matrix G is positive semidefinite, so $\det G \geqslant 0$.

Hint 2. For $m = 2$, $\det G = \|v_1\|^2\|v_2\|^2 - \langle v_1, v_2\rangle^2 \geqslant 0$ is the Cauchy–Bunyakovsky inequality, with equality iff v_1, v_2 are dependent.

Hint 3. Expressing the v_i in an orthonormal basis of their span, $G = V^T V$ and $\det G = (\det V)^2$ when the dimension is m: $\det G$ is the squared m-dimensional volume of the parallelepiped spanned by the v_i (zero iff they are dependent).

Problem 10

Hint 1. All four integrals are positive, so clear the denominators: you need $2\int_0^1 xf^2 \cdot \int_0^1 f < \int_0^1 f^2 \cdot \int_0^1 (f^2+1)$. Compare the factors separately: which of $\int xf^2$ and $\int f^2$ is larger, and which of $2\int f$ and $\int(1+f^2)$?

Hint 2. $\int_0^1 xf^2 \leqslant \int_0^1 f^2$ because $x \leqslant 1$, and $1 + f^2 - 2f = (1-f)^2 > 0$ everywhere, so $\int(1+f^2) > 2\int f$ (a positive Riemann integrable function has positive integral). Multiply, using $\int f^2 > 0$ and $\int f > 0$.